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@PuckerLab

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Prof. Dr. Boas Pucker

Long-Read Plant Genomics



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Availability of slides

- All materials are freely available (CC BY) - after the lectures:
  - GitHub: https://github.com/bpucker/LRG

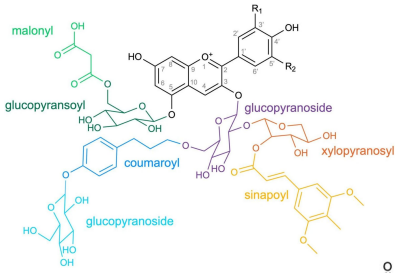
- Questions: Feel free to ask at any time

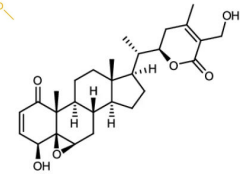
- Feedback, comments, or questions: pucker[a]uni-bonn.de

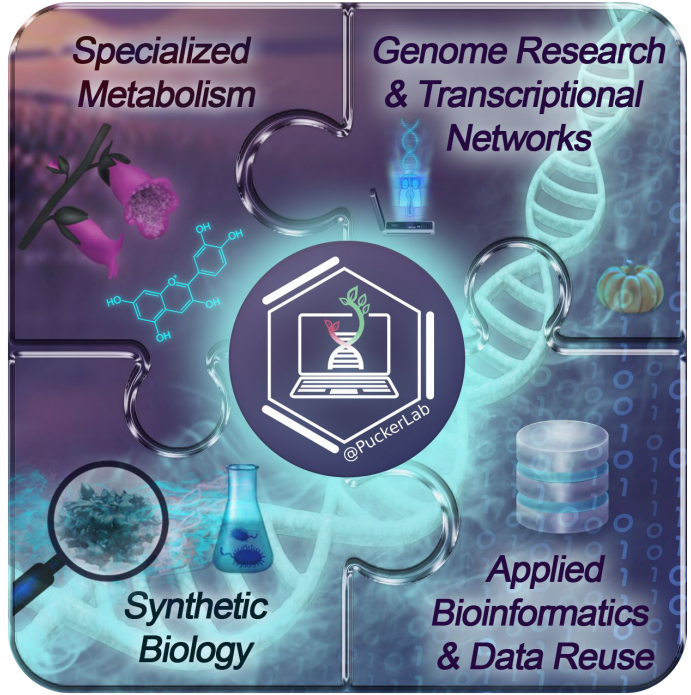


My figures and content can be re-used in accordance with CC BY 4.0, but this might not apply to all images/logos.

Plant Biotechnology and Bioinformatics





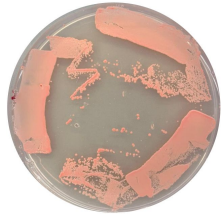


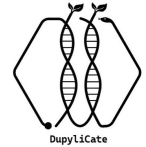














Wolff et al., 2023: 10.1515/jib-2023-0033
Dassow et al., 2025: 10.1101/2025.01.31.635922
Grünig et al., 2025: 10.1186/s12870-025-08006-3

How to sequence a plant genome?

Fresh plant material without legal restrictions





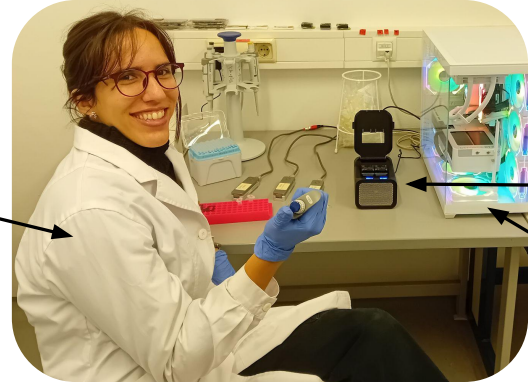
Plant DNA extraction and preparation for ONT sequencing

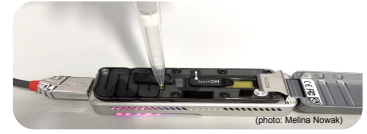
DOI
dx.doi.org/10.17504/protocols.io.bcvyiw7w
Boas Pucker[1]
[1]Plant Biotechnology and Bioinformatics, Institute of Plant Biology, TU Braunschweig

MAR 21, 2020

Established DNA extraction protocol

Highly skilled personnel







High tech devices



Extensive practice

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Cloud computing resources

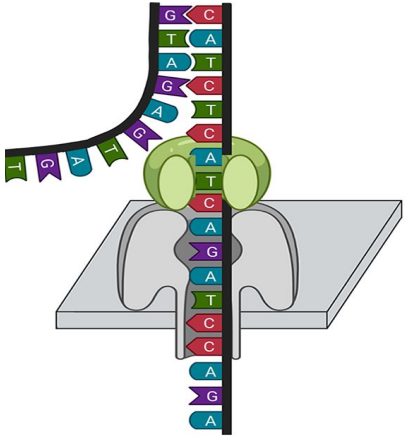
MinKNOW

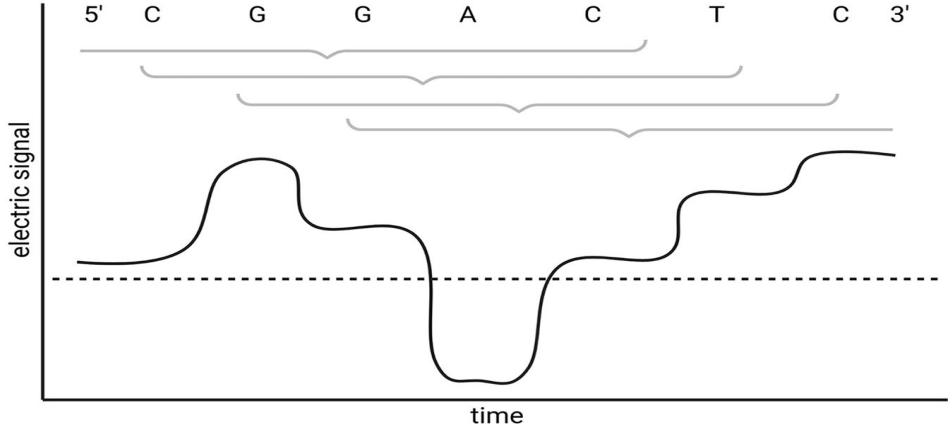
Sophisticated software

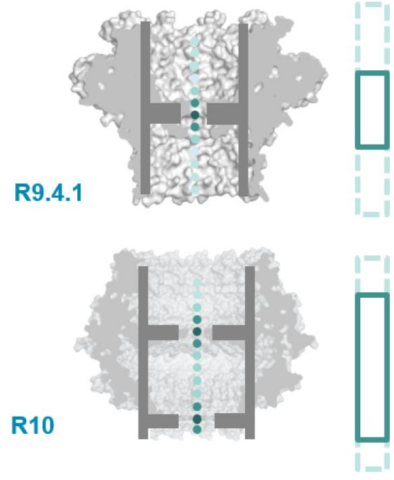
Oxford Nanopore Technologies (ONT)

Under development since the 1980s

“Analysis of anything, anywhere by anyone” … not restricted to DNA sequencing







Pucker et al., 2022: 10.1017/qpb.2021.18
https://nanoporetech.com/about-us/news/r103-newest-nanopore-high-accuracy-nanopore-sequencing-now-available-store

ONT sequencing devices



MinION



PromethION 2 Solo

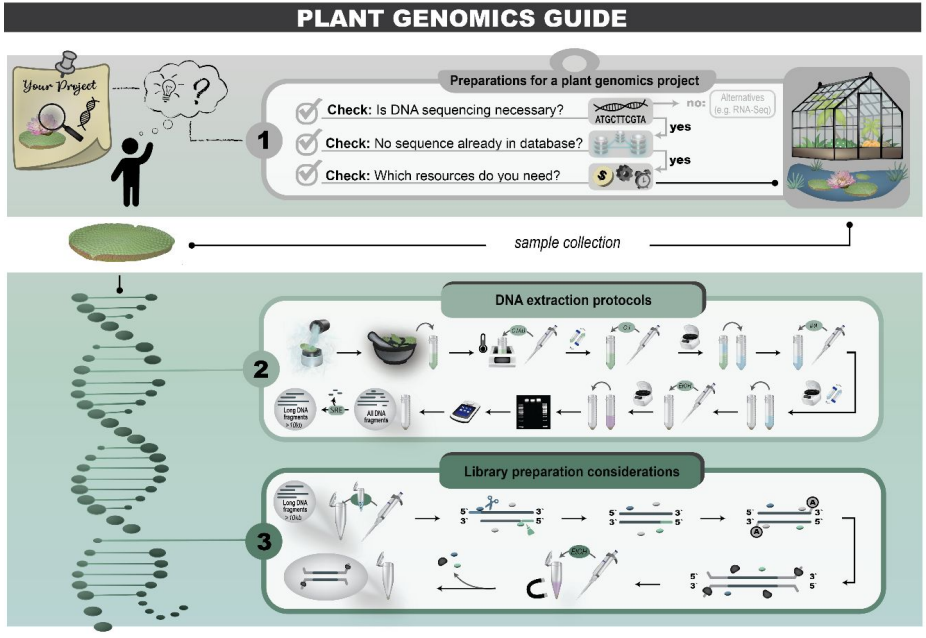
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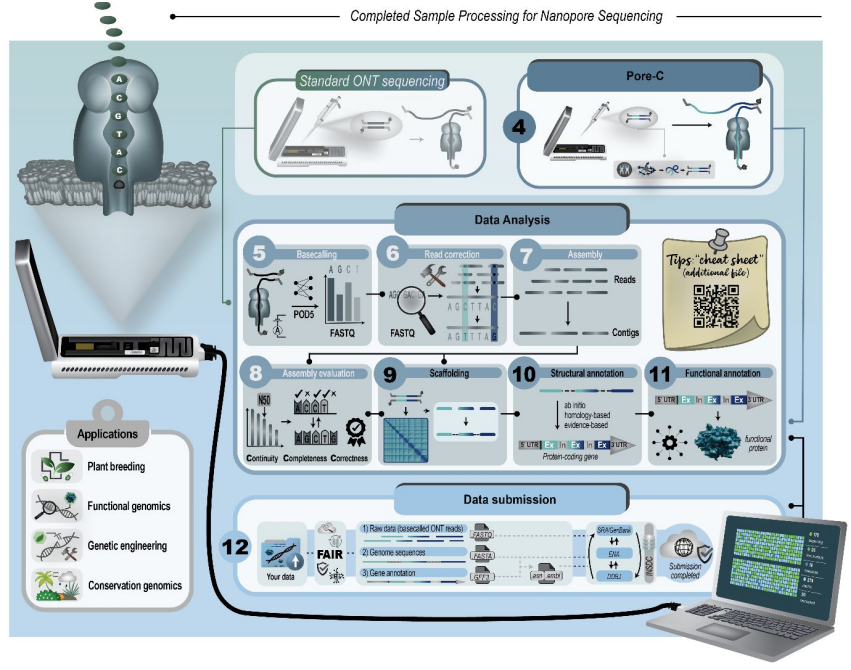
Flow cells: MinION vs. PromethION



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Established workflow for plant genomics



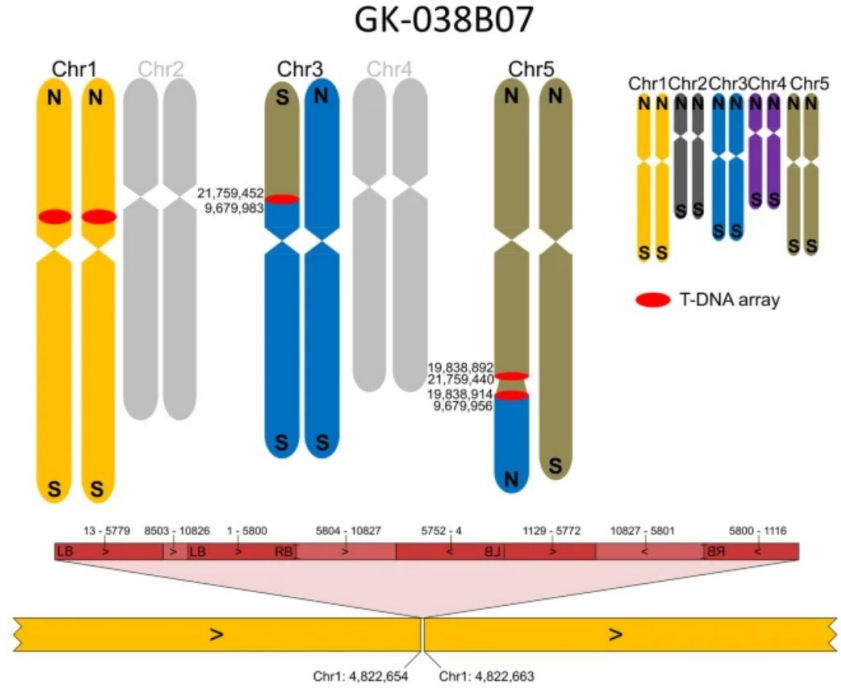


Supplementary file with all commands!

de Oliveira et al., 2026: 10.1186/s12864-026-12623-z

Long-read genotyping of T-DNA lines

- T-DNA insertions generate knock-outs
- T-DNA insertion lines were backbone of gene function elucidation in *A. thaliana*
- GABI-Kat comprises >80,000 lines
- Genotyping with ONT sequencing reveals off-targets and number of T-DNAs

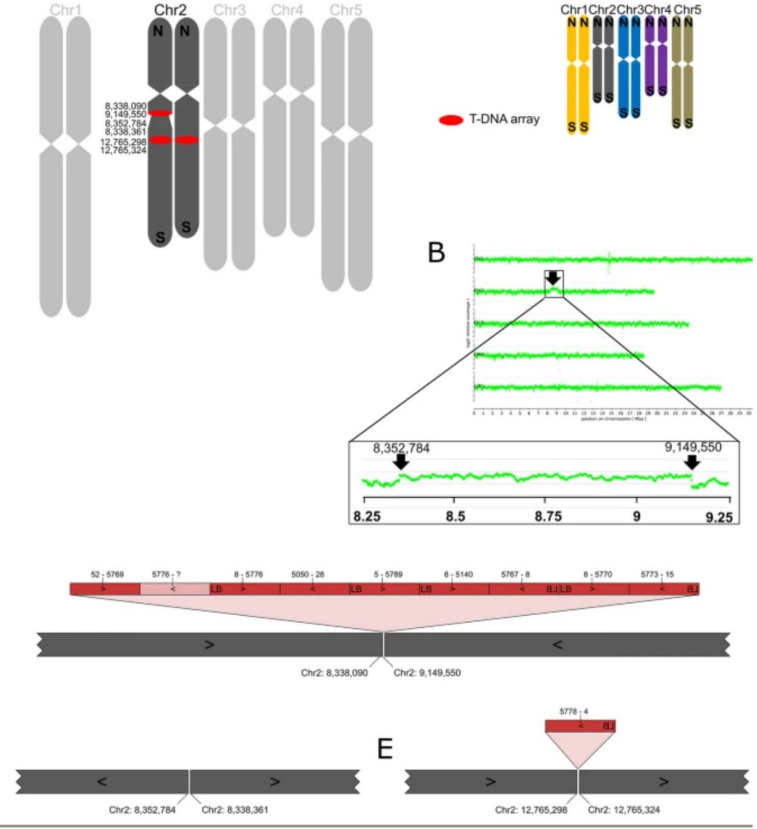


Pucker et al., 2021: 10.1186/s12864-021-07877-8

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GK-767D12

- Photosynthesis is a fundamental property of plants
- Central genes of the photosynthesis cannot be knocked-out (would be lethal)
- GK-767D12 has a structural variant potentially affecting AT2G21385/CGLD11 (Chloroplast Grana-Less Development 11)



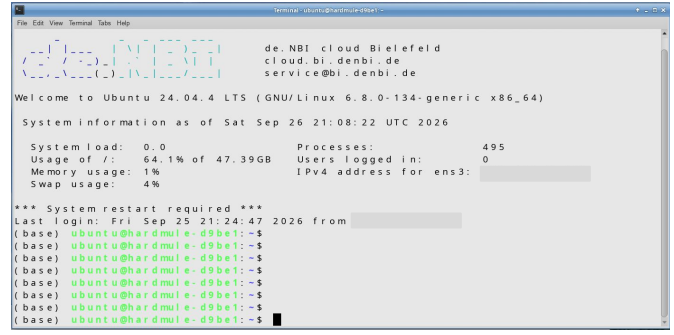
Pucker et al., 2021: 10.1186/s12864-021-07877-8

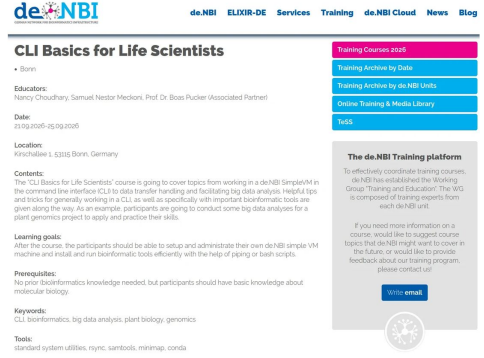
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GERMAN NETWORK FOR BIOINFORMATICS INFRASTRUCTURE

- Cloud computing resources for bioinformatics
- Hosting of services (e.g. pbb-tools.de)
- Training courses for a diversity of methods
- Network for exchange about latest developments in computational biology





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Storing sequencing data in FASTQ

- Standard format for sequencing data
- Four lines per entry:
  - Header starts with @ (title + description)
  - Sequence
  - + (optional repetition of header)
  - Quality (phred encoded in ASCII character)
- Different versions existed (outdated today)
- Example:

@seq1

ACGTACGTACGT

+

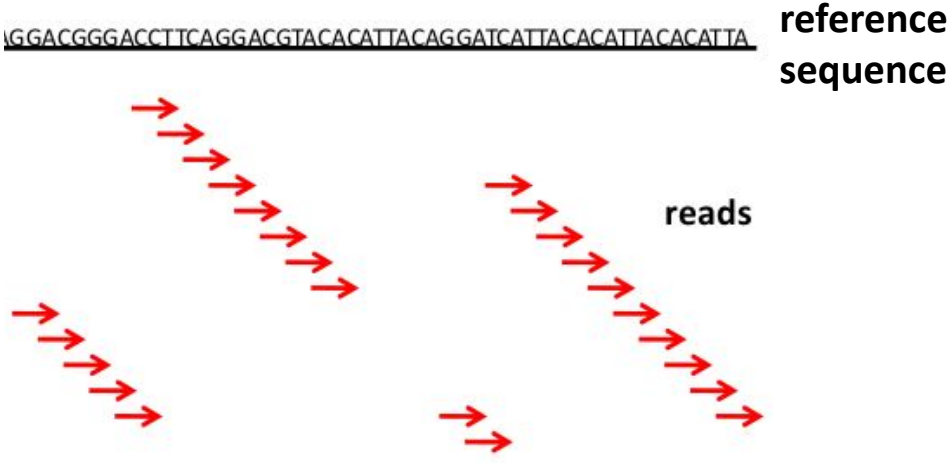
“”?CB””:DC”

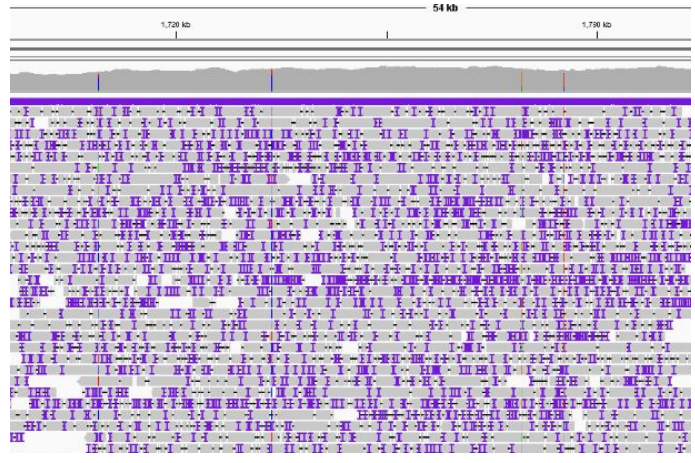
Symbol	Phred Quality Score	Probability of Incorrect Base Call
!	0	1.000
"	1	0.794
#	2	0.631
$	3	0.501
%	4	0.398
&	5	0.316

Phred quality score	Error probability	Accuracy
10	1 in 10	90%
20	1 in 100	99%
30	1 in 1,000	99.9%
40	1 in 10,000	99.99%
50	1 in 100,000	99.999%

Read mapping: concept & reality

- Alignment of reads against a reference sequence
- SAM = Sequence Alignment/Map
- BAM = Binary Alignment Map





Integrative Genomics Viewer (IGV)

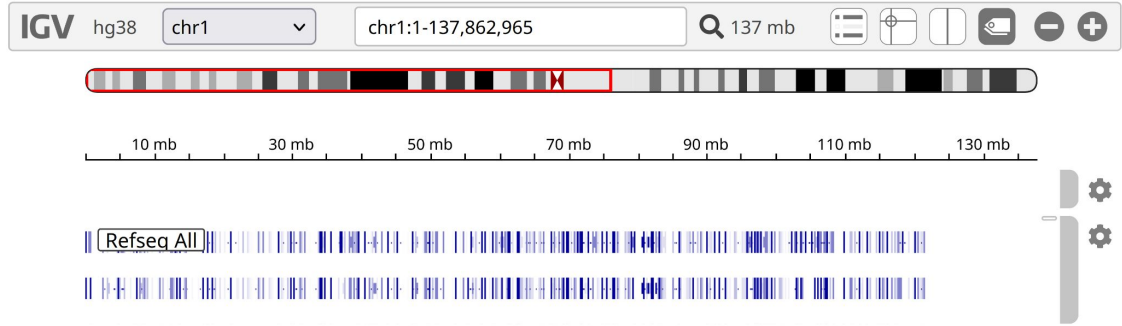
- Download IGV from:

  https://igv.org/doc/desktop/

- or use IGV online:

  https://igv.org/app/

- Load genome from file
- Load file (select BAM file)
- Load file (select GFF file)



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Plan for today

- Connect to a virtual machine (environment for computational analyses)
- Caclulate FASTQ file statistics
- Map reads against a reference genome sequence (TAIR10)
- Download the resulting BAM file
- Inspect your read mapping with IGV

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Starting VM

- Obtain a private key file (matching to public key file in VM)
- Download the README file
- Follow instructions in the README file



https://uni-bonn.sciebo.de/s/cEc5eDsYQPgDRL7

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Exercises

Please find the commands for all exercises here:

https://github.com/bpucker/LRG/blob/main/README_EXERCISES1.md

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Backup: obtaining results

- Download:
  - FASTA + FAI file
  - GFF file
  - BAM + BAI file

- Visualize read mapping in IGV

https://uni-bonn.sciebo.de/s/xb9NXnn4Z8gaTyb

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IGV instructions

- Load genome
- Load from file: GFF, BAM
- Search for gene of interest

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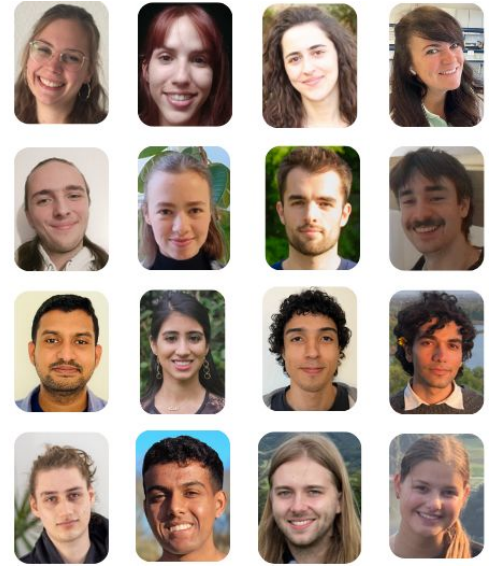
Erasmus+

- Opportunity for exchange within the EU
- Lecturer mobility
- Student mobility e.g. research visit or semester abroad

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Acknowledgements





Supported by:

de.NBI GERMAN NETWORK FOR BIOINFORMATICS INFRASTRUCTURE

Bundesministerium für Bildung und Forschung

Stiftung Innovation in der Hochschullehre

DFG Deutsche Forschungsgemeinschaft German Research Foundation

sdw Stiftung der Deutschen Wirtschaft Wir stiften Chancen!

Botanical Gardens in Braunschweig and Bonn